

Problem

The step response of a parallel RLC circuit is

$$v = 10 + 20e^{-300t} (\cos 400t - 2 \sin 400t) \text{ V}, t \geq 0$$

when the inductor is 50 mH. Find R and C .

Step-by-step solution

Step 1 of 4

Consider the step response of the parallel RLC circuit for $t > 0$.

$$v(t) = 10 + 20e^{-300t} (\cos 400t - 2 \sin 400t) \text{ V}$$

Consider,

$$u(t) \xleftrightarrow{LT} \frac{1}{s}$$

$$e^{-at} \cos(bt) u(t) \xleftrightarrow{LT} \frac{s+a}{(s+a)^2 + b^2}$$

$$e^{-at} \sin(bt) u(t) \xleftrightarrow{LT} \frac{b}{(s+a)^2 + b^2}$$

Step 2 of 4

Apply Laplace transform for $v(t)$.

$$\begin{aligned} V(s) &= \frac{10}{s} + 20 \left[\frac{s+300}{(s+300)^2 + 400^2} \right] - 40 \left[\frac{400}{(s+300)^2 + 400^2} \right] \\ &= \frac{10}{s} + \frac{20s+6000-16000}{s^2+600s+250000} \\ &= \left[\frac{10}{s} + \frac{20s-10000}{s^2+600s+250000} \right] \\ &= \frac{10(s^2+600s+250000)+20s^2-10000s}{s(s^2+600s+250000)} \end{aligned}$$

Consider the characteristic equation for $V(s)$.

$$s(s^2 + 600s + 250000) = 0$$

$$s^2 + 600s + 250000 = 0 \dots\dots (1)$$

Step 3 of 4

Consider the characteristic equation for parallel RLC circuit.

$$s^2 + \frac{1}{RC}s + \frac{1}{LC} = 0 \dots\dots (2)$$

Compare equation (1) with equation (2).

Compare constant term.

$$\frac{1}{LC} = 250000 \dots\dots (3)$$

Determine the value of capacitor C .

Substitute 50 mH for L in equation 3.

$$\frac{1}{(50 \times 10^{-3})C} = 250000$$

$$C = \frac{1}{(50 \times 10^{-3})250000}$$

$$C = 80 \times 10^{-6} \text{ F}$$

Therefore, the value of the capacitor C is $\boxed{80 \times 10^{-6} \text{ F}}$.

Step 4 of 4

Determine the value of resistor R .

Compare s terms.

$$\frac{1}{RC} = 600 \dots\dots (4)$$

Substitute 80 μF for C in equation 4.

$$\frac{1}{R(80 \times 10^{-6})} = 600$$

$$R = \frac{1}{(80 \times 10^{-6})600}$$

$$R = 20.83 \Omega$$

Therefore, the value of the resistor is $\boxed{20.83 \Omega}$.